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MILITARY SPECIFICATION

SEMITRAILERS, VAN: MODIFIED COMMERCIAL

This specification is approved for use within the US Army Tank-Automotive Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The specification covers modified commercial van type semitrailers. Vehicles procured under this specification are commercial items that are required to be warranted by the manufacturer as specified in acquisition documents.

1.2 Classification. The semitrailer shall be one of the following types, classes and sizes, as specified (see 6.2). Permissible combinations of type, class and size shall be as shown in table I.

Type I - Semitrailer, van, furniture
Type II - Cancelled
Type III - Semitrailer, van, refrigerator
Type IV - Semitrailer, van, closed
Type V - Semitrailer, van, open
Type VI - Semitrailer, van, side curtain.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2330

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Class A - 6350 kilograms (kg) (14,000 pounds), single axle
Class B - 8150 kg (18,000 pounds), single axle
Class C - 10 900 kg (24,000 pounds), single axle
Class D - 18 150 kg (40,000 pounds), tandem axle
Class E - 22 700 kg (50,000 pounds), tandem axle
Class F - 31 700 kg (70,000 pounds), tandem axle.

Size 28 - Cancelled
Size 30 - 9140 millimeter (mm) (30 foot) length
Size 32 - Cancelled
Size 35 - 10 670 mm (35 foot) length
Size 40 - 12 190 mm (40 foot) length
Size 45 - 13 720 mm (45 foot) length
Size 48 - 14 630 mm (48 foot) length.

TABLE I. *Permissible combinations*.

Type	Class	Size				
		30	35	40	45 1/	48 1/
I	C			X		
III	C D	X		 X		
IV	C D F	X	 X	 X	 X	 X
V	C D	X		 X		
VI	E					X

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

1/ May not be fully highway legal in some States.

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SPECIFICATIONS
FEDERAL

CCC-C-419	- Cloth, Duck, Unbleached, Plied Yarns, Army and Numbered.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-H-3912	- Hardwood Components, Fabricated: For Military Vehicles.
MIL-C-20696	- Cloth, Coated Polyester or Nylon, Waterproof.
MIL-T-25959	- Tiedowns, Cargo Aircraft.

STANDARDS
FEDERAL

FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
MIL-STD-1595	- Qualification of Aircraft, Missile and Aerospace Fusion Welders.
MS 75021	- Connector, Receptacle, Electrical - 12-Contact, Intervehicular, 28-Volt, Waterproof.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF DEFENSE
Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

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DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations (FMCSR).
Federal Motor Vehicle Safety Standards (FMVSS).

(Application for copies of DoT publications should reference the Code of Federal Regulations, 49 CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

* OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)
Code of Federal Regulations, 29 CFR.

(Application for copies of OSHA publications should reference the Code of Federal Regulations, 29 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
ASME Boiler and Pressure Vessel Code (BPVC)
Section IX - Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies of ASME publications should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)
AWS B2.1 - Standard for Welding Procedure and Performance Qualifications.
AWS D1.1 - Structural Welding Code-Steel.

(Application for copies of AWS publications should be addressed to the American Welding Society, 2501 NW 7th Street, Miami, FL 31025.)

THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual.

(Application for copies of ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG1	- Motors and Generators.

(Application for copies of NEMA publications should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, D.C. 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70	- National Electrical Code.

(Application for copies of NFPA publications should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SAE, INC.

SAE Standards and Recommended Practices

J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DoD adopted).
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
J700	- Upper Coupler Kingpin - Commercial Trailers and Semitrailers.
J702	- Brake and Electrical Connection Locations - Truck Tractor and Truck Trailer.
J1292	- Automobile, Truck, Truck-Tractor, Trailer and Motor Coach Wiring.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.

Year Book.

(Application for copies of Tire and Rim Association publications should be addressed to The Tire and Rim Association, Inc., 175 Montrose West Ave., Copley, OH 44321.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.11, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be the manufacturer's standard or optional items, sold commercially to the public, which meet or exceed the requirements of this specification. All vehicle items shall be as represented and rated in the vehicle and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components and shall be available to the engineering offices of the procuring activity, prior to delivery of the items. The vehicle shall be a standard commercial semitrailer, of a model sold to a significant number of buyers other than the Government, in the course of normal business operations.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. As specified by the procuring activity for the appropriate military service (see 6.2), the exterior color shall be in accordance with MIL-STD-1223.

3.1.1.2 Marking and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification marking and data plates shall be in accordance with MIL-STD-1223.

3.1.1.3 Rustproofing. The vehicle shall be rustproofed in accordance with FED-STD-297.

3.1.1.4 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the semitrailer manufacturer's standard matching hardware.

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3.1.1.5 Welders and welding. In addition to the requirements of 3.18.4, all welders employed in the fabrication of the semitrailer shall be certified, before any welding is accomplished, in accordance with MIL-STD-1595; the welding qualifications of the ASME BPVC, Section IX; or AWS B2.1 or D1.1. The certification that the welders have passed the qualifications test shall be on file at the contractor's facility and shall be available for review by the Government.

3.1.1.6 Lubrication chart. An illustrated or schematic diagram type lubrication chart shall be provided on the semitrailer. The chart shall direct attention to all lubrication fittings and specify the range and grade of lubricants required for critical temperatures. The chart shall be permanently attached to the semitrailer in a readily visible location. The chart shall be inscribed on a plate conforming to composition A (class 1 or 2) or composition C of MIL-P-514.

3.1.1.7 Hardwood. Wood used in the platform of the semitrailer shall be apitong, white or red oak, or white ash as specified in MIL-H-3912. Apitong is allowed for flooring only, provided the average weight shall be not less than 753 kilograms per cubic meter (kg/m^3) (47 pounds per cubic foot (pcf)), with none of the individual pieces weighing less than 673 kg/m^3 (42 pcf), with no more than 15 percent moisture content at time of weighing. Flooring is to be Government source inspected.

3.1.1.8 Wood treatment. Wood, including apitong, shall be treated with wood preservative in accordance with MIL-STD-1223.

3.1.1.9 24-volt lighting.

3.1.1.9.1 Independent 24-volt dc system. When specified (see 6.2), in addition to the standard 12-volt direct current (dc) lighting system (see 3.13 through 3.13.3), a 24-volt dc lighting system, independent of the 12-volt dc lighting system, shall be provided. Separate wiring and bulbs shall be provided for the 24-volt dc system. The system shall conform to 3.13. The 24-volt dc system shall also conform to the Federal Motor Carrier Safety Regulations referenced in 3.13.1, except the 24-volt dc stop and turn lamps shall be combined.

3.1.1.9.2 Interconnected 24-volt dc system. When specified (see 6.2), capability of operating the normal 12-volt dc system from a towing vehicle equipped with a nominal 24-volt dc electrical system shall be provided.

3.1.1.9.3 Receptacle for 24-volt dc system. When either an independent or interconnected 24-volt dc system is specified, the front of the semitrailer shall also be equipped with a 12-contact receptacle and cover conforming to MS 75021, part number MS 75021-1. The receptacle shall be located in accordance with SAE J702. The 24-volt dc, 12-contact receptacle shall be connected to the lights as follows:

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Contact B. Connect to left-hand turn signal and stop lamp (yellow)
Contact D. Connect to ground (white)
Contact E. Connect to clearance, side marker, identification, and tail lamps (black and brown)
Contact J. Connect to right-hand turn signal and stop lamp (green).
Contact L. Connect to ground (white).

The remaining contacts shall not be connected. Circuits B and J on tactical (military design) trucks are combination stop and turn indicator circuits. On the interconnected 24-volt dc system, the normal 12-volt dc turn signal lights will function both as turn signals and stop lights, and the normal 12-volt dc stop lights will not be operational when the semitrailer is connected to a towing vehicle with a 24-volt dc power supply. Because of this condition, the stop light (red) circuit is not connected to the 24-volt, 12-contact receptacle.

3.1.1.9.4 Resistors. When an interconnected 24-volt dc system is specified, the 12-contact receptacle shall be provided with resistance in each circuit to reduce the voltage of a tactical towing vehicle from a nominal 28 volts dc to 12 volts dc. Each circuit resistor shall be selected to reduce the voltage of the tactical truck to within the maximum rated voltage of the semitrailer electrical components. The resistor assembly shall be located in a protective housing and provided with adequate ventilation or a heat sink to prevent overheating and any heat damage to resistors, wiring, or adjacent components.

3.1.1.10 Lifting and tiedown attachments. The semitrailer shall be equipped with lifting and tying down attachments. Lifting and tying down attachments shall conform to type II or type III of MIL-STD-209. A transportation plate conforming to composition A (class 1 or 2) or composition C of MIL-P-514 shall be attached to the semitrailer. The transportation plate shall be inscribed with a diagram showing the lifting attachments and proposed lifting slings, the capacity of lifting attachments and proposed lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the semitrailer, showing the center of gravity, shall be provided on the transportation plate. Tiedown attachments shall be identified by stenciling or other suitable markings. Tiedown markings shall clearly indicate that the attachments are intended for the tiedown of the semitrailer on the carrier when shipped.

3.1.1.11 Cargo tiedown rings. When specified (see 6.2), floor type cargo tiedown rings shall be furnished. Tiedown rings shall be flush mounted in the floor and shall accept MB-1 tiedown devices as specified in MIL-T-25959. Tiedown ring installation shall use existing semitrailer structure to the maximum extent possible. Each tiedown ring and its attachment shall be capable of withstanding not less than a 910 kg (2,000 pound) pull in the vertical direction and at a 45 degree angle to the vertical direction. Tiedown rings shall be corrosion-resistant. Tiedown rings shall be located in two longitudinal rows, 610 mm (24 inches) on either

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side of the semitrailer centerline. Each row shall consist of tiedown rings, symmetrically spaced, with not more than 910 (36 inches) longitudinal distance between rings.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle shall comply with all Federal Motor Vehicle Safety Standards applicable to the type of semitrailer furnished and in effect on the date of manufacture.

3.2.2 OSHA and FMCSR regulations. The vehicle, with all furnished attachments, accessories and equipment, shall enable compliance by any user with all OSHA and FMCSR user regulations that are vehicle or vehicle equipment dependent. Regulations shall be those that are applicable to a commercial user of such a similar type, class and size of semitrailer for the same general use. See 6.1.

3.2.3 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be available to the appropriate engineering office of the procuring activity.

3.2.4 Dissimilar metals. All dissimilar metals used throughout the vehicle shall be insulated from one another to prevent galvanic or electrolytic action.

3.2.5 Prohibited materials. Asbestos materials shall not be used in any form in any part of the vehicle. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram or activity per item equals or exceeds 0.01 microcuries.

3.2.6 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

3.2.7 Lubrication. Lubrication means shall be provided for all parts of the semitrailer normally requiring lubrication. Where the use of high lubricating pressure will damage grease seals or other parts, fittings with pressure release shall be used.

3.2.8 Safety. All equipment or exposed portions of the equipment which are subject to extreme temperatures and inclement weather and all rotating or reciprocating parts which are of such a nature or so located as to become a hazard to operating personnel shall be insulated, fully enclosed, or properly guarded.

3.2.9 Electromagnetic radiation. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551.

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3.3 Performance. The semitrailer shall evidence no part failure, deformation, permanent set or interference between parts when towed, both empty and when loaded as specified in 3.4:

(a) At speeds as great as 32 kilometers per hour (km/h) (20 miles per hour (mph)) over unimproved roads and reasonably hard uneven terrain;
(b) At speeds as great as 97 km/h (60 mph) over improved roads; and
(c) At speeds as great as 16 km/h (10 mph) on 10 percent side slopes on unimproved roads and reasonably hard uneven terrain.

3.3.1 Turning ability. The semitrailer shall be capable of assuming a 90 degree angle to a coupled truck tractor without cramping and without damaging the semitrailer or the truck tractor.

3.3.2 Tracking ability. The semitrailer shall conform to the tracking requirements of Federal Motor Carrier Safety Regulation 393.70(a) when operated under the conditions specified in 3.3.

3.3.3 Brake performance. The service brakes shall stop the truck tractor-semitrailer combination, with the semitrailer loaded as specified in 3.4, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52. Requirements shall be met with a truck tractor loaded to its full rated gross vehicle weight (GVW) and gross combination weight (GCW).

3.4 Payload and payload distribution. The rated payload capacity, evenly distributed over the length of the platform, shall be not less than that specified in table II.

TABLE II. Payload.

Type	Class	Payload rating (not less than) kg (pounds)
I Furniture van	C	10 900 (24,000)
III Refrigerator van	C D	10 900 (24,000) 18 150 (40,000)
IV Closed van	C D F	10 900 (24,000) 18 150 (40,000) 31 700 (70,000)
V Open van	C D	10 900 (24,000) 18 150 (40,000)
VI Side curtain van	E	22 700 (50,000)

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3.4.1 Total axle load. The axle(s) shall be positioned so that the total load imposed on the tires, measured at the ground, with the semitrailer loaded as specified in 3.4, shall be not more than 9070 kg (20,000 pounds) for single axle semitrailers and 15 400 kg (34,000 pounds) for tandem axle semitrailers.

3.5 Dimensions and clearances. The semitrailer, uncoupled from the towing vehicle, resting level on the landing legs, without payload, on level ground, shall conform to the dimensions specified in table III.

TABLE III. Dimensions and clearances.

Overall length (not less than)	As specified for applicable size in 1.2.
Overall width of body (not more than)	2440 mm (96 inches).
Width over tires (not more than)	2540 mm (100 inches).
Overall height (not more than)	3810 mm (150 inches).
Ground clearance (not less than)	230 mm (9 inches).
Clearance under landing gear (not less than)	280 mm (11 inches).
Landing gear swing clearance (radius from centerline of kingpin to any portion of semitrailer 150 mm (6 inches) or more below upper fifth wheel plate) (not less than)	2130 mm (84 inches).
Distance from centerline of kingpin to front end of semi-trailer (not less than)	610 mm (24 inches) for single axle semitrailers. 910 mm (36 inches) for tandem axle semitrailers.
Swing radius from centerline of kingpin to most distant point on semitrailer nose (not more than)	1330 mm (52.5 inches) for single axle semitrailers. 1470 mm (58 inches) for tandem axle semitrailers.
Floor height	790 mm (31 inches) for type I; 1420 mm (58 inches) for type III; 1320 mm (54 inches) for types IV, V and VI.
Distance from centerline of single axle or rearmost tandem axle to end of semitrailer (not more than)	1520 mm (60 inches).
Upper fifth wheel height	1220 ±25 mm (48 ±1 inch) for single axle semitrailers. 1330 ±25 mm (51 ±1 inch) for tandem axle semitrailers.

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3.6 Construction and stresses. The body construction shall be stressed-skin (monocoque or semi-monocoque). The exterior skin, consisting of roof, sides and end sheets shall be of aluminum alloy. When exterior sheets are furnished without formed corrugations, the sheets shall be stretcher-leveled. Materials shall be free from any defects and imperfections that may affect the performance and appearance of the vehicle. Aluminum alloys shall have corrosion-resistant properties inherent in the chemical composition of the aluminum. Either interior or exterior side wall posts shall be used. Rear body members shall be steel. Otherwise, body structural members (other than the skin) may be steel or aluminum alloy. Doors, body, roof and panel joints shall be watertight and leakproof. All doors and other cutouts shall be reinforced. When the semitrailer is loaded as specified in 3.4, the stress in compression members shall not exceed 1/2 of the critical Euler stress = (SCR) = ($\pi^2 E(r/l)^2$); 2/3 of the material yield strength; or 1/3 of the material ultimate strength, whichever is less. Under the above cited conditions of loading, the deflection below horizontal shall be not greater than 50 mm (2 inches). The unloaded semitrailer may have a maximum of 50 mm (2 inches) of upward camber over the length of the platform.

3.6.1 Stress analysis. A stress analysis for the semitrailer, loaded as specified in 3.4, shall be performed. The stress analysis shall include shear and moment diagrams and deflection calculations. Stress calculations shall include complete analysis of the floor, crossmembers and longitudinal members.

3.6.2 Roof panels. Except for type V open van, the roof shall be of one piece and of stretcher level aluminum with roof bows on not more than 610 mm (24-inch) centers. The roof panel shall be secured to the roof bows to prevent panel vibration. The roof shall be capable of withstanding a static load of not less than 1.3 kg on any square meter of area (30 pounds on any square foot) without permanent deformation. The roof shall be crowned to enhance drainage. A drip molding or rain gutter shall be provided over all doors.

3.6.3 Front end. The front end shall have beveled or rounded corners which sacrifice a minimum of interior cube. The front end shall conform to the swing radius dimension in table III. The front end structure shall conform to FMCSR 393.106, including the capability of withstanding a horizontal static load equal to 0.4 times the payload capacity of the semitrailer.

3.7 Axles. Class A, B and C semitrailers shall have a single rear axle. Class D, E and F semitrailers shall have tandem rear axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with the semitrailer loaded as specified in 3.4. The wheel bearings and axle spindles shall be oil lubricated. The oil viscosity shall be in accordance with the manufacturer's recommendations. The hubcaps shall have a window for visual determination of oil level. Provision for venting or another method of withstanding internal pressure buildup, and for replenishing the oil supply, shall be incorporated.

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3.7.1 Sliding axle assembly. When specified for size 35 and longer semitrailers (see 6.2), a sliding axle, spring and tire assembly shall be furnished. The sliding axle assembly shall have an adjustment length of not less than 2440 mm (96 inches) in increments of not more than 150 mm (6 inches). The arrangement shall permit the assembly to be manually unlocked, slid to a new position beneath the semitrailer by using the towing vehicle with the semitrailer brakes locked, and then manually relocking the assembly in the new position. The assembly shall be rated for the loads imposed with the semitrailer loaded as specified in 3.4.

3.8 Suspension system. Each component of the suspension system shall have a rated capacity at least equal to the load imposed, measured at the ground, with the semitrailer loaded as specified in 3.4. Clearances shall prevent interference between the tires and any other parts of the semitrailer under the operating conditions specified herein.

3.8.1 Full air suspension. When specified (see 6.2), the semitrailer shall be equipped with a full air suspension system, with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the semitrailer loaded as specified in 3.4. Hydraulic double-acting shock absorbers shall be provided on the axle assembly. The suspension shall be provided with a mechanism at each wheel to assure lifting of the wheel and axle when jacking the semitrailer from the applicable jacking location. The suspension system shall incorporate leveling valves with time delay to avoid constant air consumption. Shock absorbers and springs shall be fabricated of materials capable of operating in ambient -46 degrees Celsius (°C) (-50 degrees Fahrenheit (°F)) to +52°C (+125°F). The suspension system shall have none of the sprung weight carried on anything other than the air springs. No other spring medium, such as leaf, coil, or rubber springs, shall be used in conjunction with the air springs.

3.9 Wheels, rims and tires. The semitrailer shall have dual disc type wheels on each axle. The rims shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Multi-piece rims shall not be furnished. Tire and rim sizes shall be the same for all wheels on the semitrailer.

3.9.1 Tires. Tires shall be steel belted radial, tubeless type, with highway tread. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the semitrailer loaded as specified in 3.4. Tires and tire ratings shall conform to Tire and Rim Association or to European Tyre and Rim Technical Organisation recommendations.

3.9.2 Spare tire assembly. The semitrailer shall have an inflated spare tire mounted on a spare wheel or rim, installed in a carrier. The spare tire shall be of the same size, tread design and load range (ply rating) as the tires in use on the semitrailer.

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3.9.3 Spare tire carrier. A spare tire carrier shall be installed under the semitrailer, on the curbside, in an accessible location. Means shall be provided for securing the tire within the carrier to prevent pilferage and accidental loss. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material. The carrier shall enable the safe removal and installation of the spare tire assembly from and to the carrier and ground without personnel positioning themselves or any part of their body under the spare tire assembly.

3.10 Brakes. The brakes shall be of the full air type and shall conform to Federal Motor Carrier Safety Regulations 393.40, 393.42, 393.43 and 393.45 through 393.47. Brake linings shall be of non-asbestos material. The braking system shall include:

(a) Automatic slack adjusters
(b) Piping
(c) Hose connections
(d) Glad hands
(e) Retracting rubber diaphragm or spring loaded dust covers or dummy glad hands equipped with security chains or cables
(f) Provisions for stowing dummy glad hands and
(g) All other components required for a complete air-brake system.

Glad hands shall conform to SAE J318. Glad hand location shall comply with SAE J702. The braking system shall be installed in a manner which provides road clearance for travel over uneven terrain and protection against damage caused by objects striking components. No part of the braking system shall extend below the bottom of wheel rims.

3.10.1 Parking brakes. Spring or air diaphragm mechanical lock type parking brakes shall be provided. The parking brakes shall be automatically applied upon disconnection of the supply (emergency) air line and under emergency braking conditions. The parking brakes shall hold the semitrailer, with specified payload, on a 10 percent grade despite the depletion of the compressed air supply. The parking brake shall hold the loaded semitrailer headed either up or down the grade. Parking brakes shall conform to Federal Motor Carrier Safety Regulation 393.41.

3.11 Upper fifth wheel plate. The upper fifth wheel plate shall be designed for completely covering and for use with either a full oscillating or fore and aft rocking lower fifth wheel. The upper fifth wheel plate shall be of sufficient strength to be fully supported by a lower fifth wheel 910 mm (36 inches) in diameter and shall conform to Federal Motor Carrier Safety Regulation 393.70(b). The forward end of the upper fifth wheel plate shall have a turned up lip not less than 50 mm (2 inches) high for ease of coupling. The kingpin shall be of heat treated alloy steel and shall conform to SAE J700.

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3.12 Landing gear. The semitrailer shall have two vertical lift, telescopic, nonrotating landing legs, with two speed gears and a handcrank on the curbside. Landing legs shall be equipped with not less than 230 mm (9 inch) diameter steel wheels or self-leveling sand shoes of not less than 250 mm by 250 mm (10 inch by 10 inch) square bearing area. Supports for the crank extension shaft and clips for holding the crank when folded shall be provided. The landing gear shall withstand, without deformation, the combined static and dynamic forces due to the proportion of the gross weight sustained and the forces resulting from impact during coupling and uncoupling operations. When placed in travel position, the landing gear legs shall remain positively locked. The landing gear shall be protected to prevent the entrance of foreign matter which would impair its functioning or mechanical efficiency. The maximum cranking effort to raise the fully loaded semitrailer shall be not more than 45 kg (100 pounds).

3.12.1 Landing gear range of adjustment. The landing gear shall have a range of adjustment sufficient to vary the height of the upper fifth wheel from 1190 mm (47 inches) to not less than 1320 mm (52 inches) from the ground. With the semitrailer coupled to a truck tractor and in level position, the clearance under the fully retracted landing gear shall be not less than as specified in table III.

3.12.2 Air/hydraulic powered landing gear. When specified (see 6.2), in addition to the requirements of 3.12 and 3.12.1 (less hand crank, etc.), the landing gear shall be air/hydraulic powered. The landing gear system shall include a control box equipped with a 12-volt dc motor driven pump, master control valve, leveling valve, booster, main oil tank, air side oil tank, key switch and mounting hardware.

3.13 Lighting. All lights and reflectors shall be protected from operational hazards by mounting in recessed or otherwise guarded locations. Lights and reflectors shall not be mounted on vertical surfaces of the rub rail (unless recessed and fully protected) or on the semitrailer bumper. All electrical wiring shall conform to SAE J1292. Clearance, side marker and identification lamps shall be constructed for easy removal and replacement of lamps and lenses without the use of any tools.

3.13.1 Twelve-volt dc system. A 12-volt dc lighting system conforming to 3.13, 3.13.2 and 3.13.3 shall be provided. The 12-volt dc lighting system shall conform to Federal Motor Carrier Safety Regulations 393.9, 393.11, 393.20, 393.22, 393.23, 393.25 through 393.29, 393.32 and 393.33.

3.13.2 Receptacle for 12-volt dc system. The front of the semitrailer shall be equipped with a 7-contact receptacle conforming to SAE J560 with the connectors connected and color coded as specified therein. The receptacle shall be provided with a spring loaded cover. The receptacle shall be located in accordance with SAE J702.

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3.13.3 Domelights. Except for type V open van, ceiling mounted domelights spaced on not more than 2620 mm (8 foot) centers for the length of the semitrailer shall be provided. All interior lights shall be controlled by a switch located on the rear of the curbside wall, recessed to prevent damage by cargo. The switch shall be accessible to an operator, standing on the ground at the curbside of the semitrailer. When side door(s) are specified, the switch shall be of the 3-way type, with a second 3-way recessed switch located near the side door, accessible from the ground.

3.14 Rear end protection. Semitrailer rear end protection shall be in accordance with Federal Motor Carrier Safety Regulation 393.86. A dock bumper, extending not less than 38 mm (1.5 inches) beyond the rear of semitrailer, shall be furnished. The dock bumper shall be the full width of the semitrailer and shall be located approximately at floor level.

3.15 Mud flaps. Mud flaps shall be provided to the rear of the rear wheels, in accordance with SAE J682. A metal strip not less than 3 mm (0.125 inch) thick and not less than 25 mm (one inch) wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 25 mm (1 inch) high by 32 mm (1.25 inch) wide by 2.4 mm (0.094 inch) thick shall be furnished at each bolt.

3.16 Semitrailer types.

3.16.1 Type I furniture van. The furniture van semitrailer shall be fully enclosed and shall have double rear doors and double curbside doors. The inside body dimensions shall be as follows:

(a) Width, not less than 2340 mm (92 inches)
(b) Height, ahead of drop, not less than 2390 mm (94 inches)
(c) Height, behind drop, not less than 2920 mm (115 inches).

3.16.1.1 Furniture van platform. The platform shall be of the low floor height, drop type, with wheel housings. Rub rails shall extend the full length of the semitrailer. The floor shall be not less than 28 mm (1-1/8 inch) thick hardwood or not less than 26.6 mm (1-1/16 inch) thick laminated hardwood, both in lengths of not less than 1830 mm (6 foot). Hardboard shall conform to 3.1.1.7 and 3.1.1.8. Boards shall be laid lengthwise with tongue-and-groove or shiplap joints. Flooring shall be secured with countersunk phosphate-coated or other corrosion-resistant screws or bolts. Not less than two securing screws or bolts shall be installed at each butt cross sill and not less than two on each side of each butt joint. The top of countersunk screw or bolt heads shall be not less than 3.2 mm (1/8 inch) and not more than 6.4 mm (1/4 inch) below the top of the wood surface. Butt joints shall be centered over crossmembers and shall be staggered. Rails or pockets for attaching loading ramps (see 3.16.1.5), shall be provided at the rear and at the side door.

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3.16.1.2 Furniture van interior. Interior panels shall be attached to the wall and ceiling members. Panels shall be of exterior grade AD plywood not less than 6.4 mm (1/4-inch) thick. Interior radius corners may be aluminum lined. Not less than four rope cleats or rope tiepins shall be provided on each sidewall post back of the drop and not less than three on each post forward of the drop. Cleats or tiepins shall be recessed and shall not protrude beyond the lining. Rug rods or straps shall be furnished in accordance with the manufacturer's standard practice.

3.16.1.3 Furniture van doors. Unless roll-up doors are specified (see 3.16.1.4), doors shall be hung between box posts with not less than three heavy-duty pin strap hinges on each door, permitting the doors to fold back against the sides of the body. Doors shall be provided with double cam latches and an operating handle. The rear door opening shall be not less than 2720 mm high by 224 mm wide (107 inches high by 88 inches wide). Double curbside doors, covering a door opening not less than 2130 mm (84 inches) wide and not less than 2440 mm (96 inches) high, shall be furnished. Means shall be provided to hold all doors in the full open position.

3.16.1.4 Roll-up rear and side doors. When specified (see 6.2), roll-up overhead rear and curbside doors shall be provided. The rear door shall provide a clear opening of not less than 2240 mm (88 inches) in width and not less than 2720 mm (107 inches) in height. The curbside door shall provide a clear opening of not less than 2130 mm (84 inches) in width and not less than 2440 mm (96 inches) in height. The doors shall be of the sectional type having not less than six sections. The door sections shall incorporate joints of the tapered tongue and groove or shiplap type. The doors shall be plastic-coated or aluminum-covered 19 mm (3/4 inch) marine plywood construction. Door section corrosion-resistant hinges shall be mounted on the inside of the door, one on each side of each section joint. The door tracks shall be of the corrosion-resistant type equipped with a positive stop at the end of the tracks. Door rollers, counter-balance units and cables of corrosion-resistant construction shall be furnished. The doors shall be weathertight and shall close to not less than 6.4 mm (1/4 inch) below floor level. Heavy duty corrosion-resistant cam-operated lever-type locks shall be provided, equipped with provisions for use of padlocks. Heavy duty corrosion-resistant closed type grab handles shall be provided on the left outside bottom of the doors. Nylon double loop pull down straps not less than 50 mm (2 inches) wide and 300 mm (12 inches) in length shall be provided and located adjacent to the door locks.

3.16.1.5 Furniture van loading ramps. Two loading ramps of all metal construction, complete with hooks and handles shall be furnished. Each ramp shall be approximately 1020 mm (40 inches) wide. When in place with the upper end in the door rails or pockets and the lower end on the ground, the ramping angle shall be not more than 15°. Each ramp shall be capable of rigidly supporting a concentrated load of 454 kg (1,000 pounds) without permanent deformation, when used in either a ramping or a bridging position.

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3.16.1.6 Furniture van optional moving equipment. When specified (see 6.2), the following optional moving equipment shall be furnished in the quantities and in accordance with the requirements specified by the purchaser:

(a) Furniture pads
(b) Appliance truck
(c) Walkboards
(d) Piano board with straps
(e) Floor runners, canvas
(f) Refrigerator cover
(g) Stove cover
(h) Tie webbing (roll)
(i) Piano straps.

3.16.1.7 Furniture van underslung cabinets. Underslung stowage cabinets, extending the full width of the semitrailer and, generally, from just behind the gooseneck to just ahead of the rear tires, shall be provided. The cabinets shall be provided with hinged access doors on both sides of the semitrailer.

3.16.2 Cancelled.

3.16.3 Type III refrigerator van. The refrigerator van semitrailer shall be fully enclosed and insulated, and shall have a continuous straight floor without wheelhousings or kickup. The interior and exterior panels shall be straight with no bulges resulting from construction or the addition of insulation. Double rear doors shall provide a full width opening for access into the body. The inside body dimensions shall be as follows:

(a) Width, not less than 2180 mm (86 inches)
(b) Height, not less than 2240 mm (88 inches).

3.16.3.1 Refrigeration performance. The refrigeration unit shall have sufficient cooling capacity to reduce the air temperature inside the van body from 38°C (+100°F (Fahrenheit) to not more than -18°C (0°F) in not more than 8 hours, while the semitrailer is in an ambient air temperature of not less than 38°C (+100°F). The refrigeration unit shall maintain -18°C (0°F) inside the van body under all normal operating conditions, including road speeds of 88 kilometers per hour (km/h) (55 mph), in any ambient air temperature from -18°C (0°F) to not less than 38°C (+100°F). The interior temperature shall be thermostatically controlled by means of a calibrated, manually adjustable device, and the refrigeration unit shall maintain, within (±3°C) (±6°F), any selected temperature from 2°C (+35°F) to -18°C (+0°F) when the ambient air temperature is between 2°C (+35°F) and 38°C (+100°F). The temperatures throughout the van body interior from one point to any other point shall not vary more than 1.5°C (2-1/2°F). The refrigerator unit shall have a heating unit or

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cycle, capable of maintaining the van body interior at 2°C (35°F) when the ambient air temperature is -32°C (-25°F) while the semitrailer is operating at a road speed of 88 km/h (55 mph).

3.16.3.2 Refrigerator van insulation. Body insulation shall be nonhygroscopic, self-extinguishing, cellular material and shall be resistant to growth of fungus and retention of vermin. The thermal conductivity shall not exceed 2.16 watts per square meter per °C per centimeter ($w/m^2/°C/cm$) (0.15 Btu per square foot per °F per inch) of thickness at a mean temperature of 21°C (10°F). Insulation shall be sealed and bonded at all joints and voids. Insulation shall be anchored to prevent sag or insulation voids.

3.16.3.3 Refrigerator van inner panels. Inner panels shall be of fiberglass. Fiberglass panels shall be not less than 2.5 mm (0.100-inch) thick. Panels shall be attached to wall structural members with fasteners spaced on not more than 610 mm (2 foot) centers. The inside area of wall and doors shall be covered from the floor to a height of not less than 610 mm (2 feet) above the floor with not less than 3.2 mm (1/8-inch) thick fiberglass secured with phosphate coated or other corrosion-resistant screws.

3.16.3.4 Refrigerator van platform. The platform shall be flat, without wheelhousings or kickup. The platform shall be extruded aluminum alloy having longitudinal corrugated or T-shaped sections not less than 50 mm (2 inches) in height and spaced on centers not greater than 64 mm (2-1/2 inches). The platform shall be capable of withstanding concentrated loads imposed by forklifts having 6350 kg (14,000 pound) axle load supported by 180 mm (7-inch) wide tires spaced 760 mm (30 inches) apart on the tire centerline. The sections, with corrugations or T-sections lengthwise of the body, shall be welded together to prevent leakage of moisture through the floor. The sides and front of the floor shall have a metal flashing extending upwards for a distance of not less than 75 mm (3 inches) and shall be sealed between metal and interior insulation to minimize sidewall condensation. The floor design shall permit semitrailer operation at speeds up to and including 88 km/h (55 mph), over highways, with rated payload. Floor insulation shall be placed over body crossmembers and shall be sealed from moisture by an under pan. An under pan of 1 mm (0.040-inch) thick aluminum, 6.4 mm (1/4-inch) thick exterior grade AC plywood, 1.5 mm (0.0598-inch (U.S. revised standard gage No. 16)) thick steel, or 0.6 mm (0.0239-inch) thick galvanized steel on plywood shall be furnished. Not less than four self-sealing water drains shall be provided, two at front and two at rear of the platform.

3.16.3.5 Refrigerator van rear doors. Unless a roll-up rear door is specified (see 3.16.3.6), two doors, the full width of the semitrailer and not less than 2110 mm (83 inches) in height, shall provide a clear opening for access into the body at the rear. Doors shall be insulated so that body insulating efficiency will not be reduced. Door gaskets shall be installed

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to ensure a positive seal. Double cam latches, with an operating handle, shall be installed on each door to provide positive closing action. The door locking device shall have provisions for use of a padlock. Doors shall be equipped with an inside handle operated lock release and shall be capable of being opened from inside when padlocked on the outside.

3.16.3.6 Refrigerator van roll-up overhead rear door. When specified (see 6.2), a roll-up overhead rear door shall be provided. The door opening shall be within 125 mm (5 inches) of the full width and within 230 mm (9 inches) of the full height of the body interior. The door shall be of the sectional type having not less than five sections. Door sections shall be not less than 29 mm (1.125 inch) thick pultruded fiberglass filled with polyurethane foam with molded plastic end caps or equal. Door insulation shall maintain body insulating efficiency. Door gaskets shall be installed and shall be designed to ensure a positive seal. A door locking device, with operating handle, shall be installed to provide positive closing action and shall have provisions for the use of a padlock. The door shall be capable of being opened from inside when locked. Door section corrosion-resistant hinges shall be mounted on the inside of the door, not less than four hinges at each section joint. Door track shall be the manufacturer's standard corrosion-resistant type equipped with a positive stop at the end of the track. Door rollers, counterbalance unit, and cables of corrosion-resistant construction shall be the manufacturer's standard. The door shall be weathertight. One heavy duty corrosion-resistant grab handle closed type, shall be provided on the outside bottom of the door. One nylon double-loop pull-down strap, not less than 38 mm (1.5 inches) wide and 300 mm (12 inches) in length, shall be provided and located adjacent to the door lock.

3.16.3.7 Refrigerator van side door. When specified (see 6.2), the body shall be equipped with a curbside swing type door. The door shall be not less than 1850 mm (73 inches) in height and shall provide not less than a 760 mm (30-inch) clear width opening. The door shall be insulated so that body insulating efficiency will not be reduced and shall be furnished with gaskets to ensure a positive seal. A door locking device, with operating handle and provision for use of a padlock, shall be installed. The door shall have provision for securing in the open position.

3.16.3.8 Refrigerator van stiff leg supports. The size 32 and shorter refrigerator van semitrailer shall be provided with two stiff leg supports mounted at the front end. Stiff leg supports shall be of the pivot or telescoping type. The stiff legs shall pivot or extend from travel position to support position and shall be retained in place with locking pins. The legs shall be adjustable for length. The legs shall be adequate to prevent the semitrailer, uncoupled from a truck tractor, from nosing over, while the semitrailer is initially loaded at the front by forklifts as described in 6.1.

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3.16.3.9 Refrigerator van interior lights. The domelights required by 3.13.3 shall be moistureproof and corrosion-resistant and shall be provided with glass and or plastic lenses. Lamps shall be 12 volt dc potential, wired to operate from the refrigerating unit batteries. A circuit opening switch controlling all interior lights shall be operated by closing the door.

3.16.3.10 Refrigeration unit. The refrigeration unit shall contain all components, equipment, and accessories normally furnished on the manufacturer's standard unit for comparable commercial applications and shall include the components specified herein. The refrigerator capacity shall be equal to or shall exceed the refrigeration performance requirements of 3.16.3.1. The unit shall have all thermometer and controls mounted at the front of the body exterior. The refrigeration unit shall be either the single unit type or of the split unit type. The split unit may be either of two configurations:

(a) Refrigeration unit (evaporator, condenser, compressor) in a single compartment installed in the nose of the semitrailer and a separate engine-generator power unit mounted under semitrailer; or
(b) An evaporator section mounted in the nose of the semitrailer and a separate engine-condensing unit mounted under the semitrailer.

Applicable components of the refrigeration unit shall be mounted on a frame attached to the front of the van exterior. The unit shall incorporate lifting eyes to facilitate removal and installation. The unit shall be enclosed with either quickly removeable or hinged metal panels capable of protecting the refrigeration unit under any precipitation conditions, of up to 75 mm (3 inches) of rain per hour impinging at any angle from the vertical to 45° off vertical toward the unit. The unit shall extend into, but occupy a minimum of cubage within the van. The opening shall be reinforced and tightly sealed. All components and assemblies shall withstand vibrations encountered in normal vehicle operation. The refrigerant shall be:

(1) R_{22}-chlorodifluoromethane; or
(2) R_{502} [azeotrope of R_{22}-chlorodifluoromethane ($CHClF_2$) and R_{115}-chloropentafluoroethane (C_2ClF_5)]; or
(3) R_{134A}-tetrafluoroethane (CH_2FCF_3).

The refrigeration unit shall be furnished with a diesel engine and with an auxiliary electric motor. Means shall be provided for automatically switching the drive from the engine to the electric motor as electrical power is switched on by means of of a manually operated switch, leaving the engine disconnected. The electric motor shall conform to NEMA MG1 and shall be rated for 208 volt, 3 phase, 4 wire, 60 Hertz, alternating current operation. The refrigeration unit shall include components and controls as follows:

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(a) Fuel tank, integrally or separately mounted, sufficient for 12 hours of operation; and a fuel gage, visible from outside the unit enclosure
(b) Starting motor
(c) Generator or alternator. Maintenance-free storage batteries and battery container, integral or separately mounted.
(d) Pressure unloading device on compressor or centrifugal engine clutch
(e) Automatic defrosting effected by reverse cycle, electric or by bypassing hot refrigerant from the discharge line to a built-in heater in the evaporator.
(f) A replaceable liquid line refrigerant filter/dehydrator using renewable dehydrator.
(g) A thermostatic expansion valve in the liquid line.
(h) A heat exchanger in the suction line.
(i) A digital or dial thermometer to indicate body interior temperature, readable at a 1520 mm (5-foot) distance outside of the unit.
(j) A refrigerant compound suction-pressure gage facing the outside of the unit enclosure
(k) Manually set thermostatic controls for maintaining the body interior at the temperatures indicated in 3.16.3.1.
(l) Manually operated control switch which will shut down the engine and start the electric motor. This switch shall prevent the engine from being accidentally started while the electric motor is in operation, powered by an outside electrical source.
(m) Controls shall include an automatic start and stop operation or shall be of a modulating type.

3.16.3.11 Refrigerator van receptacle and caution plate. A weather-resistant slave receptacle shall be provided on the refrigeration unit, in an accessible, convenient location for connection to a 208 volt, 3 phase, 4 wire, 60 Hertz power source. Wiring shall be in accordance with NFPA 70. The receptacle shall be 3 wire, 4 pole, polarized type with male contacts. The receptacle shall be rated for use with 250 volt ac and shall be of adequate capacity for the motor furnished. A mating female receptacle connector shall be provided. The receptacle shall have a spring hinged cover, except when the receptacle is facing in a downward direction. An aluminum or corrosion-resistant plate with stamped letters, that will be legible in the event of being painted over, shall be mounted near the slave receptacle. The plate shall not be less than 64 by 100 mm (2-1/2 by 4 inches) in size, bearing the following inscription in characters of easily read size:

"Power source required: 208 volt, 3 phase, 4 wire, 60 Hertz, ac.
Notice: Power source of 110 volt or 220 volt, single phase, will not operate this unit."

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3.16.3.12 Refrigerator van air leakage. The air leakage from the semitrailer shall not exceed 340 liters per minute (L/m) (12 cubic feet per minute) at an air pressure differential of 6.4 mm (0.25 inches) of water, without external air movement.

3.16.3.13 Refrigerator van operating instructions. The refrigeration unit shall be equipped with instruction plate(s) or decal(s) mounted inside a removable panel of the unit enclosure. The panel shall have appropriate lettering on the outside, directing attention to the location of the operating instructions. The operating instructions shall describe any special or important procedures to be followed in operating and servicing the unit. Metal plate(s) shall conform to MIL-P-514, composition A (class 1 or 2) or composition C, and shall be permanently affixed to the panel with screws or bolts of not less than 3.2 mm (1/8 inch) diameter. Printed decals shall have a pressure sensitive adhesive backing.

3.16.4 Type IV closed van. Type IV van semitrailer shall be fully enclosed and shall have a continuous straight floor without wheelhousings or kickup. The inside body dimensions shall be as follows:

(a) Width, not less than 2340 mm (92 inches)
(b) Height, not less than 2390 mm (94 inches).

3.16.4.1 Platform. The platform floor shall be of either the composite or laminated hardwood type, conforming to 3.1.1.7 and 3.1.1.8. The platform shall be capable of withstanding concentrated loads imposed by forklifts having a 6350 kg (14,000 pound) axle load supported by 200 mm (8-inch) wide tires spaced 760 mm (30 inches) apart on the tire centerline. Composite floors shall have hardwood floorboards not less than 32 mm (1-1/4 inch) thick finished dimensions and not less than 1830 mm (6 feet) long, accurately fitted and laid lengthwise between hat-shaped channels constructed of not less than 1.9 mm (0.0747-inch). Floorboards shall be countersunk for, and secured with, phosphate-coated or other corrosion-resistant countersunk head screws, two at each cross sill, spaced on centers of not more than 300 mm (12-inches), and two on each side of each butt joint. Butt joints shall be centered over crossmembers and shall be staggered. Laminated hardwood floors shall be not less than 27 mm (1-1/16 inch) thick finished dimension, supported on cross sills spaced on centers of not more than 300 mm (12 inches) or 35 mm (1-3/8 inch) finish dimension supported on cross sills spaced on centers of not more than 410 mm (16 inches). Rub rails shall extend the full length of both sides of the platform.

3.16.4.2 Closed van interior panels. Interior panels shall be of not less than 6.4 mm (1/4-inch) exterior grade AD plywood treated in accordance with 3.1.1.8. Interior panels shall be attached to wall structural members with fasteners spaced on not more than 300 mm (1-foot) centers. The inside areas of all walls and doors shall be covered to a height of not less than 300 mm (2 feet) above the floor with not less than 12 gage (2.657 mm) (0.1046 inch) thick sheet steel scuff plates. Sheets shall be treated or coated to

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resist corrosion and need not be painted. The scuff plate shall be installed with corrosion-resistant countersunk screws, rivets, or low profile round head screws. The scuff plate shall be screwed, riveted or welded to formed corner plates.

3.16.4.3 Closed van doors. Unless a roll-up rear door is specified (see 3.16.4.4), double rear doors shall be provided. The clear door opening shall be not less than 2240 mm (88 inches) in width and not less than 2240 mm (88 inches) in height. Doors shall be hung with not less than three heavy-duty pin strap hinges on each door. Doors shall permit folding back against the body sides and shall be provided with means for holding the doors in the full open position. Doors shall be provided with gaskets and shall be tightly fitted to ensure a seal against the entrance of water and dust. Each door shall be provided with double cam latches with an operating handle and shall have provisions for locking with a padlock.

3.16.4.4 Closed van roll-up rear door. When specified (see 6.2), a roll-up rear door shall be provided. The door shall provide a clear opening not less than 2240 mm (88 inches) in width and not less than 2240 mm (88 inches) in height. Otherwise, the roll-up rear door shall conform to 3.16.1.4.

3.16.4.5 Closed van side door. When specified (see 6.2), a curbside swing type door, not less than 1520 mm (60 inches) wide by 1830 mm (72 inches) high, shall be furnished. The door shall be located near the center of the van body. The door shall be of aluminum faced plywood or of all aluminum with bracing to prevent sagging. The door shall be equipped with at least three hinges. A cam-type locking device shall be provided with an operating handle and shall be installed to ensure positive closing under all operating conditions. Provisions shall be incorporated for a padlock. Handles shall be located to permit operation from ground level. The door shall have a weather-tight rubber or plastic seal all around. The side wall shall be reinforced at the door opening. The body shall be equipped with latches to hold the door in a fully open position.

3.16.4.6 Closed van vent doors. The semitrailer shall be equipped with four vent doors. Two doors shall be located on the front wall and two on the rear. The doors shall be not less than 150 by 300 mm (6 by 12 inches) on the front and not less than 300 by 300 mm (12 by 12 inches) on the rear. The doors shall have screen protectors.

3.16.4.7 Closed van loading ramp. When specified (see 6.2), a loading ramp, not less than 4270 mm (14 feet) in length shall be furnished. The ramp shall be capable of rigidly supporting a concentrated load of not less than 410 kg (900 pounds) without permanent deformation. The ramp shall be stored in its own storage frame positioned between the frame rails, accessible from the rear of the semitrailer. The unit to be similar or equal to Melcher Model EASY-14A.

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3.16.4.8 Closed van interior rope ties. Unless cargo control racks are specified (see 3.16.4.9), interior rope ties, either fold-down lashing rings, rope cleats, or rod type, shall be installed and secured to wall frame members. Rope ties shall be arranged in three evenly spaced tiers, spaced on not more than 1250 mm (4 foot) longitudinal centers on each wall, including the front wall. The first tier shall be located just above the scuff plate, the second tier at 1/3 of the wall height, and the third tier at 2/3 of the wall height. Rope ties shall project not more than 9.5 mm (0.375 inch) into the cargo area beyond the wall lining. Rope tie protrusions shall be rounded and constructed to prevent injury to personnel and snagging of payload inside the cargo area.

3.16.4.9 Closed van interior cargo control tracks. When specified (see 6.2), interior cargo control tracks shall be installed. Tracks shall be attached to side posts through full length side post wood fillers. Tracks shall be attached between posts through full length wood fillers, to the side wall lining, on not more than 150 mm (6 inch) centers. The cargo control tracks shall be mounted in two tiers, horizontally on each sidewall. The first tier shall be located just above the scuff plate, the second tier at 2/3 of the wall height. Each cargo control track shall be the full length of the van body interior. The track fitting holes or slots shall be on approximately 64 mm (2-1/2 inch) centers. The cargo control tracks shall be fabricated of steel with a thickness of not less than 12 gage (2.657 mm) (1.046 inch). Two cargo control track bars shall be provided. The tracks and cargo control bars shall be capable of withstanding a rearward static load of 17.8 kN (4,000 pounds) without permanent deformation.

3.16.5 Type V open van. The open van semitrailer shall conform to requirements of 3.16.4 through 3.16.4.3, except as modified herein. The inside wall height shall be not less than 2240 mm (88 inches) and the door opening height with header installed shall be not less than 2080 mm (82 inches). A removeable door header shall be furnished. Laminated hardwood is not acceptable for the flooring.

3.16.5.1 Open van tarpaulin, bows and tiedown devices. Removeable tarpaulin bows, with two longitudinal top straps or ropes, shall be furnished. The tarpaulin bows shall be of steel or aluminum, mounted in sockets on the cap rail, and spaced at nominal 910 mm (36-inch) intervals from the rear door header to the front of the semitrailer. Top straps or ropes shall be provided with anchoring devices and means for tension adjustment. Tarpaulin tiedown devices shall be installed at a height not to exceed 1830 mm (6 feet) from the ground.

3.16.5.2 Open van tarpaulin. A fitted tarpaulin shall be provided. The tarpaulin color shall harmonize with the vehicle exterior color. The tarpaulin shall be fabricated of hard texture cotton duck cloth not less by weight per square yard than No. 10 cotton duck conforming to type I of CCC-C-419; or of vinyl coated nylon conforming to type II, class 2 of

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MIL-C-20696. The cloth shall be treated or shall be inherently resistant to fire, water, weather, and mildew. All seams shall be double stitched. The tarpaulin shall cover the entire body when spread over a load level with the top of the van walls and shall extend down the sides, front, and rear to within 1220 mm (4 feet) of the platform. Reinforcing pads of the size, shape, and in the positions required to protect the tarpaulin at the wall corners shall be sewn in the tarpaulin.

3.16.5.3 Tiedown ropes. Manila tiedown ropes, 9.5 mm (3/8 inch) in diameter, shall be spliced through grommets spaced not more than 610 mm (2 feet) apart. The free ends of the rope shall be secured with metal clips or seized with waxed twine to prevent raveling. Tiedown ropes shall be of sufficient length to permit tying to the tiedown devices when the edges of the tarpaulin are 1830 mm (6 feet) above the platform.

3.16.6 Type VI side curtain van. The side curtain van shall have a continuous straight floor without wheelhousings or kickup and shall be fully enclosed as specified herein. The inside body dimensions shall be:

(a) Width, not less than 2340 mm (92 inches)
(b) Height, not less than 2390 mm (94 inches).

3.16.6.1 Curtain van curb and streetside walls. Curb and streetside walls shall be curtains in lieu of conventional aluminum sides. The curtains shall extend the full length and height of the semitrailer. The curtains shall ride on a curtain roller or cable assembly. The roller assembly shall be attached to the upper roof area and shall be continuous the full length of the semitrailer. Vertical tensioning straps not less than 44 mm (1-3/4 inch) wide shall be located on 610 mm (24 inch) centers. The tensioners shall have bottom hooks to secure the curtain on the underside of the frame rail. A ratchet assembly to provide a positive brake and to maintain curtain tension when closed and to provide a positive release shall be provided. A single operator, without any assistance, shall be capable of opening the curtain assembly. Any additional roof supports shall be removable in order to permit easy access to the loading cargo area.

3.16.6.2 Curtain van platform. The platform shall conform to 3.16.4.1.

3.16.6.3 Rear curtain door. A rear curtain shall be provided in lieu of conventional double rear doors. The rear curtain assembly shall be similar to the curb and streetside walls. A single operator without any assistance shall be capable of opening the curtain assembly.

3.16.6.4 Curtain material. The curtain material for the side and rear walls shall be made of rip-stop reinforced polyester impregnated with polyvinyl chloride (PVC) material. The color of the curtain material shall be the same as the semitrailer exterior color.

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3.17 Servicing and adjusting. Prior to acceptance of the semitrailer by the Government inspector, the contractor shall service and adjust the semitrailer for immediate operational use. The servicing and adjustment shall include at least the following:

(a) Inflation of all tires
(b) Adjustment of brakes
(c) A check for proper functioning of all lights
(d) Complete lubrication with grades of lubricants recommended for ambient air temperature at the delivery point.

3.18 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Bolted, riveted and welded construction used shall be in accordance with the highest standards of the industry.

3.18.1 Metal fabrication. Metal used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. All bends of a major character shall be made with controlled means in order to ensure uniformity of size and shape.

3.18.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.18.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.18.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. See 3.1.1.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the

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contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of the product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. The first vehicle of each type, class and size produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance with the requirements of the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.3.1 Semitrailer weight. The semitrailer shall be weighed to determine the curb weight and the distribution of the curb weight on the kingpin and the rear axles. The imposed loading on the kingpin and rear axles, with payload, shall be computed using curb weight and the semitrailer payload distributed as specified in 3.4. The rated capacity of the axles, suspension system and tires shall be compared to the loads imposed on them to determine if they are of adequate capacity to conform to contract requirements.

4.3.2 Normal road test. Unless an extended road test is specified (see 4.3.3), the first vehicle for each contract line item number (CLIN) under contract shall be road tested with and without payload. The road test shall be for the distance required in table IV, corresponding to the number of vehicles being procured under the specific contract line item number (CLIN).

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The test shall be conducted by the contractor under the direction of Government representatives. The test shall be conducted over a test course approved by the procuring activity and meeting the requirements of table V. The vehicle shall be inspected periodically during and at the completion of the test by the contractor.

TABLE IV. Road test distance.

Number of vehicles being procured	Road test distance km (miles)
1 - 10	160 (100)
11 - 20	320 (200)
21 - 30	480 (300)
31 - 40	640 (400)
41 - 50	800 (500)
51 - 100	1600 (1000)
101 - up	8050 (5000)

TABLE V. Test course requirements.

Type of road	Road test distance
Dirt	10 percent
Gravel	30 percent
Improved pavement	60 percent

4.3.2.1 Test course. The total test mileage shall be on roads and highways within the general geographic limits of the place of manufacture and adjoining states. Any special highway permits required shall be obtained by the contractor. The course shall be arranged to return the vehicle to the manufacturer's plant at the completion of each 1600 km (1,000 miles), if applicable, or at the completion of the test otherwise, for inspection and chassis servicing.

4.3.2.2 Test loads. The semitrailer shall be tested for 90 percent of the mileage with the semitrailer loaded as specified in 3.4 and for 10 percent of the mileage with the semitrailer empty. Average speeds during all road tests shall be as close as possible to those specified in 3.3.

4.3.2.3 Road test failure. The vehicle shall successfully complete the entire test. Failure of the vehicle to successfully complete the test shall be cause for nonacceptance of any of the contract quantity, pending correction of deficiencies and evidence of corrective action which shall prevent the recurrence of similar deficiencies in the contract quantity. Failure of the vehicle to successfully complete the test shall not constitute an excusable

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delay in meeting scheduled deliveries. The equipment shall be closely monitored throughout the test for interference with towing operations, and for instability as evidenced by poor tracking or uncontrollable sideslipping, skidding, swerving, or tilting. The equipment also shall be closely monitored for loosening of parts, interference between parts, leakage of fluids or lubricants, overheating of components, and for damage, permanent distortion, or excessive wear of parts and components. The equipment shall be disassembled to the extent necessary for inspection for unusual wear or damage to components. Substitution of a new semitrailer or replacement of a major component may require complete retest at the discretion of the Government representatives. Rejection of the test semitrailer shall be for damage or deficiencies, including but not limited to the following:

(a) Damage caused by collision
(b) Failure of any major component
(c) Vibration due to misalinement of wheels or frame
(d) Vibration due to the type of construction or mounting
(e) Evidence of abnormal tire wear due to misalinement or unbalance
(f) Failure of any vehicular safety device such as brakes or electrical circuits
(g) Evidence of structural weakness in any part of the vehicles, vehicle components, accessories or welds
(h) Loose mountings of parts or accessories due to workmanship or vehicular operation.

4.3.3 Extended road test. When specified (see 6.2), the first vehicle shall be subjected to an extended road test as specified in table VI. While negotiating the specified side slopes, in both directions, the equipment shall be subjected to not less than 10 sine wave turns. While negotiating the paved road course at highway speeds, the equipment shall be periodically subjected to at least 25 sudden stops at maximum braking effort and shall be periodically subjected to at least 25 left and 25 right turn rapid lane change maneuvers that simulate avoidance of collisions.

TABLE VI. Extended road test.

Surface	Load condition	Average speed km/h (mph)	Distance km (miles)
Side slope, 10 percent	Full load Half load Empty	8 - 16 (5 - 10) 8 - 16 (5 - 10) 8 - 16 (5 - 10)	
Belgian block	Full load Half load Empty	16 - 32 (10 - 20) 16 - 32 (10 - 20) 16 - 32 (10 - 20)	480 (300) 320 (200) 160 (100)

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TABLE VI. Extended road test (continued).

Surface	Load condition	Average speed km/h (mph)	Distance km (miles)
Paved road	Full load	88 (55)	3380 (2100)
	Half load	88 (55)	2250 (1400)
	Empty	88 (55)	1130 (700)
Improved gravel road	Full load	40 - 56 (25 - 35)	1450 (900)
	Half load	40 - 56 (25 - 35)	970 (600)
	Empty	40 - 56 (25 - 35)	480 (300)

4.3.3.1 Patterning. High stress areas of supporting members shall be coated with brittle lacquer prior to the road test. The semitrailer shall be fully loaded and towed over the Belgian block test course for one cycle.

4.3.3.2 Strain recording. Strain gage rosettes shall be positioned in locations of high stress concentrations. The locations shall be as determined by the brittle lacquer patterning. A minimum of ten strain gages shall be used. Locations of strain gages shall include the gooseneck, the kingpin area and transverse areas. Strains shall be recorded for each of the following three cycles of the test course:

(a) Full load 1 cycle
(b) 40 to 60 percent of full load 1 cycle
(c) No load 1 cycle

Strain/occurence histograms shall be supplied for each strain gage for each test cycle. The actual test course shall be recorded on the histogram. Pictures indicating the position of each strain gage shall be provided.

4.3.3.3 Evaluation. The ultimate, tensile and yield strengths of the gaged components shall be documented. Any strain recorded indicating stress greater than that specified in 3.6 shall be cause for rejection.

4.3.3.4 General design and performance characteristics verification. The equipment shall be verified as suitable for the intended use by satisfactorily completing the preceding inspections and tests. Equipment submitted for test shall contain all specified components and shall be properly adjusted and serviced for operation. Apparatus used in conjunction with the tests shall be calibrated for accuracy. In addition to the results of inspections and tests, test records shall include the following information, as applicable.

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(a) Name, date(s), and location of test
(b) Description of test course and weather conditions that could affect test results
(c) Condition of the equipment at the start of test
(d) Tire inflation pressure
(e) Types of lubricants used
(f) Description of test instrumentation and recording apparatus, including calibration records
(g) Description of the truck tractor used
(h) Load condition of the equipment
(i) Average and maximum speeds
(j) Distance traveled, including total test distance and accumulated distance, and the time of each major incident such as component failure or adjustment
(k) Component failure analysis and corrective measures
(l) Component adjustment, and method and ease of adjustment
(m) Any noted safety hazards
(n) Condition of the equipment at the conclusion of the test.

4.3.3.5 Reliability and maintainability report. A report pertaining to reliability and maintainability shall be furnished after completion of the extended road test. The report shall contain the following information:

(a) All failures, maintenance, and other events shall be recorded and identified by accumulated operating time, distance, cycles, or position in the test procedure as appropriate. Test conditions during the failures or irregular operations identified shall be recorded.
(b) Description of the engineering reasoning, and of any tests conducted to determine assignable causes for all failures or irregular operations identified.
(c) Description of the engineering reasoning behind any proposed corrections to be made on the production items, and the predicted effectiveness of those corrections.
(d) Test activity or contractor comments on item, features or requirements that, if modified, should improve the item.
(e) Estimates (within 5 percent of actual experienced) of manhours required for each maintenance and servicing action during the test. A brief description of the qualifications and experience of the personnel involved shall be included and shall be adequate to allow comparison to the personnel anticipated in similar field work.

4.3.4 Brake tests. Brake tests to determine conformance to requirements shall be conducted during the road test. The brake tests shall be conducted with the fully loaded semitrailer coupled to a fully loaded truck tractor. The truck tractor shall have a gross vehicle weight rating (GVWR) and gross combination weight rating (GCWR) not exceeding that specified in table VII. The truck tractor shall be without excessive brake capacity. GVWR and GCWR

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shall be the maximum ratings on the tractor only and not necessarily the actual maximum GVW and GCW of the combination of test vehicles. Failure of the vehicle combination to stop within the required 12 200 mm (40 feet) from a speed of 32 km/h (20 mph) shall be cause for rejection.

TABLE VII. Maximum tractor characteristics.

Class of semitrailer	Max. tractor GVWR, kg (pounds)	Max. tractor GCWR, kg (pounds)
A	10 900 (24,000)	18 150 (40,000)
B	12 700 (28,000)	19 960 (44,000)
C	12 700 (28,000)	22 700 (50,000)
D	17 250 (38,000)	31 800 (70,000)
E	20 200 (44,500)	36 300 (80,000)
F	22 700 (50,000)	45 360 (100,000)

4.3.5 Turning ability test. A truck-tractor shall be coupled to the semitrailer and driven through turns up to 90 degrees to the towed vehicle, right and left. At the conclusion of five turns in each direction, the tractor and the semitrailer shall be examined for evidence of binding, misalinement or other damage.

4.3.6 Parking brake test. To test parking brakes, the semitrailer, with rated payload, shall be placed on a grade, first headed up the grade and then headed down the grade, and tested to determine conformance to 3.10.1. These tests shall be conducted with the semitrailer disconnected from the towing vehicle and resting on the landing legs.

4.3.7 Landing leg test. With the semitrailer disconnected from the towing vehicle and loaded as specified in 3.4, the landing legs shall be raised and lowered to vary the height of the upper fifth wheel over the landing gear's full range, to determine conformance to 3.12 and 3.12.1. The loaded semitrailer shall be coupled and uncoupled no less than 5 times and the landing legs shall be examined for binding and or misalinement.

4.3.8 Electrical test. An electrical test shall be conducted to determine that all vehicle lights are functioning. In addition, when an innerconnected 24-volt system is furnished, the circuits of the 12-contact receptacle shall be energized with not less than 28 volts dc regulated voltage for a time sufficient to stabilize the temperature of the resistor assembly for not less than 30 minutes in an ambient temperature of not less than 25°C (77°F). After the test, the resistor assembly, including resistors, wiring and adjacent components shall be examined for evidence of overheating, deterioration or damage.

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4.3.9 Lifting and tiedown test. Lifting and tiedown provisions, markings, and the connecting structural members of the vehicle shall be inspected for conformance with MIL-STD-209. All lifting and tiedown provisions and the connecting structural members shall meet the criteria of MIL-STD-209. Test results that show the provisions do not meet the criteria of MIL-STD-209 shall constitute failure of this test.

4.3.10 Leak test. The semitrailer, except type V, shall be subjected to a leak test by spraying fresh water at a minimum rate of 50 mm (2 inches) per hour at any angle between 30° and 45° from vertical. The test shall be performed with the water directed against all four sides, the roof and the undercarriage of the semitrailer simultaneously or section by section and shall continue for not less than 15 minutes (per section). The test shall be conducted prior to the installation of the interior insulation and lining. Following the test the semitrailer shall be inspected for water leakage.

4.3.11 Type III refrigerator van air leakage. Type III refrigerator van shall be subjected to an air leakage test. A blower, manometer, and flow meter shall be connected in series to the semitrailer cargo space through one floor drain. The remaining floor drains shall be sealed. Air leakage shall be measured by feeding air through a flow meter into the semitrailer in order to maintain 6.4 mm (0.25 inch) water pressure inside the semitrailer. The air pressure differential shall be held constant for a period of 25 minutes by throttling the blower output. Five flow meter readings shall be taken at 5-minute intervals. Nonconformance to 3.16.3.12 shall be cause for rejection.

4.3.12 Type III refrigerator van pulldown and operation test. The unloaded refrigerator semitrailer shall be placed in a controlled temperature room for a period of 6 hours with the temperature maintained at 38°C (100°F) with the semitrailer doors open. The semitrailer doors shall be closed and the ambient temperature maintained at not less than 38°C (100°F). The ambient air temperatures shall be recorded at a height of not more than half of the height of the semitrailer nor less than 1520 mm (5 feet) from the ground line, while not directly exposed to the heat source. The refrigeration unit, powered by the diesel engine, shall be operated until the temperature of the semitrailer interior is reduced to -18°C (0°F) as specified in 3.16.3.1. The period of time required to obtain this temperature shall be recorded. Immediately following this cooling test, the refrigeration unit shall maintain the interior temperature of the semitrailer at the specified low value for a period of not less than 2 hours, with equal periods of operation using the engine and the electric motor. The refrigeration system instruments and controls shall be calibrated before or during the test by comparison with a certified standard temperature measuring instrument calibrated in the appropriate temperature range. During the two hour temperature maintenance period, the temperature shall be recorded at five minute intervals. Temperature readings shall be made using thermocouples calibrated at the time of the test, with a digital readout instrument, preferable with a time/temperature recorder.

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4.3.13 Type III refrigerator van cooling control test. Control of ambient air temperature will not be required provided that the ambient temperature is maintained above 2°C (+35°F) and below 38°C (100°F) during the test period. The thermostatic controls shall be set at -18°C (0°F), 07°C (20°F) and 20°C (35°F), and the inside body temperature brought to approximately -15°C (5°F) above each control setting. After the inside body temperature has reached each specified test value, the controls shall be tested, using the engine and the electric motor to operate the refrigeration unit, each for a period of not less than 30 minutes. The temperature shall be recorded during the test at five minute intervals to determine conformance to 3.16.3.1.

4.3.14 Type III refrigerator van highway cooling test. The thermostatic controls shall be set at -18°C (+0°F) and the semitrailer operated at the top highway speed for a period of not less than 30 minutes in an ambient air temperature between 2°C (+35°F) and 38°C (+100°F) to determine conformance to 3.16.3.1. When the semitrailer is traveling at 88 km/h (55 mph), proper operation of the heat cycle shall be demonstrated immediately after the highway cooling test by switching the unit to the highest heat mode to demonstrate a temperature rise on the thermometer of at least 10°C (15°F) in 30 minutes.

4.3.15 Type III refrigeration unit rain test. The refrigeration unit shall be subjected to a simulated rain storm to determine compliance with 3.16.3.10. Fresh water shall be sprayed on the refrigeration unit at a rate of 75 mm (3 inches) per hour at any angle between vertical and 45° off vertical against the exposed sides of the unit. The refrigeration unit shall be exposed to the spray for 1/2 hour while idle and then shall be started and operated for 1/2 hour while the spray continues.

4.3.16 Type VI curtain assembly test. A curtain assembly test shall be conducted to verify the conformance to 3.16.6.1 and 3.16.6.3. Without assistance to the operator, the operator shall open, then close all the curtains, and secure the assembly within 10 minutes. The total time shall not exceed 20 minutes. Failure to pass any phase of the test shall be cause for rejection.

4.3.17 Certifications. The contractor shall certify that all wood requiring treatment in accordance with MIL-STD-1223 has been so treated. When an air suspension system is furnished, the contractor shall certify the capability of the system to operate over the temperature range specified in 3.8.1.

4.3.18 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in an as new condition for the duration of the contract. Release of the first vehicle for shipment prior to completion of the contract shall be at the discretion of the local Government Quality Assurance Representative (QAR).

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4.3.19 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under the contract until corrective action has been taken.

4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting the performance requirements contained herein.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from the manufacturer's plant to the initial receiving activity, in accordance with manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The semitrailers covered by this specification are non-tactical vehicles (NTVs), intended for use by the Armed Forces for transporting general cargo. At times, cargo will be loaded on these semitrailers by forklifts having a 6350 kg (14,000 pound) axle load supported by 200 mm (8-inch) wide tires, with a tread width between wheels of 760 mm (30-inches).

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, class and size of semitrailer required (see 1.2).
(c) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Identification of appropriate military service for painting and marking (see 3.1.1.1 and 3.1.1.2).
(e) Independent 24-volt dc lighting system, if required (see 3.1.1.9.1).
(f) Interconnected 24-volt dc lighting system, if required (see 3.1.1.9.2).
(g) Cargo tiedown rings, if required (see 3.1.1.11).
(h) Stress at 33 percent of material yield strength, in lieu of 50 percent, if required (see 3.6).
(i) Sliding type axle, if required (see 3.7.1).
(j) Air suspension system, if required (see 3.8.1).
(k) Air/hydraulic powered landing gear, if required (see 3.12.2).
(l) Type I furniture van roll-up rear and side doors, in lieu of hinged, if required (see 3.16.1.4).
(m) Optional moving equipment for type I furniture van, if required (see 3.16.1.6).

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(n) Roll-up rear door, in lieu of hinged, if required for type III refrigerator van (see 3.16.3.6).
(o) Curbside hinged door, if required for type III refrigerator van (see 3.16.3.7).
(p) Roll-up rear door, in lieu of hinged, if required for type IV closed van (see 3.16.4.4).
(q) Curbside hinged door for type IV closed van semitrailer, if required (see 3.16.4.5).
(r) Loading ramp for type IV closed van semitrailer, if required (see 3.16.4.7).
(s) Interior cargo control tracks, if required in lieu of rope ties for type IV closed van (see 3.16.4.9).

6.3 Subject term (key word) listing.

Forklifts
Furniture van
Hardwood
Landing gear
Non-tactical vehicle (NTV)
Open van
Refrigerator van
Side curtain van
Upper fifth wheel
Welder's qualifications.

Custodians:
Army - AT
Navy - YD
Air Force - 99

Review activity:
Air Force - 84

User Activity:
Navy - MC

Preparing Activity:
Army - AT

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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8. PREPARING ACTIVITY

a NAME	b TELEPHONE *(Include Area Code)* (1) Commercial	(2) AUTOVON
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